

1. What is the difference between a basic filter, advanced filter, and a slicer in power BI?

ANS- Difference between Basic Filter, Advanced Filter, and Slicer in Power BI are given below:

a. Basic Filter

- A simple filter applied directly to a visual, page, or report.
- Useful for quick, straightforward filtering.

b. Advanced Filter

- Provides more complex filtering options beyond just selecting values.
- Allows conditions like greater than, less than, contains, or between.
- Example: Show products where sales are greater than ₹50,000.

c. Slicer

- A visual element added to the report canvas that acts like an interactive filter.
- Users can dynamically select values (e.g., Region, Product Category) to filter multiple visuals at once.
- More user-friendly and interactive compared to filters.

2. What is a visual level filter , and how does it differ from page-level and report- level filters?

ANS- **Visual-Level Filter**

- Applied to a single visual (chart, table, etc.).
- Only affects the data shown in that specific visual.
- Example: A bar chart showing sales by product, filtered to display only 2023 data. Other visuals on the page remain unaffected.

- Page-Level Filter

- Applied to an entire report page.
- All visuals on that page are filtered according to the condition.
- Example: Filtering a page to show only sales from the "North Region" — every chart and table on that page will reflect this filter.

- Report-Level Filter

- Applied across the entire report.
- Every page and every visual in the report is filtered by the condition.
- Example: Setting a report-level filter to show only sales from "India" ensures that all visuals across all pages display Indian sales data.

6. What is drill- down in power BI? How does it improve data exploration?

ANS- Drill-Down is a feature in Power BI that allows users to explore data at different levels of detail within a hierarchy (e.g., Year → Quarter → Month → Day).

Drill-Down Improves Data Exploration in the following manner:

- Interactive Analysis: Users can dynamically navigate through data levels instead of creating multiple charts.
- Better Insights: It reveals hidden patterns and trends that might not be visible at a summary level.
- Efficiency: Reduces clutter by combining multiple levels of data into one visual, making dashboards cleaner and more user-friendly.

10. Explain the difference between drill-down and drill-through in power BI using a real world example.

ANS-Difference Between Drill-Down and Drill-Through in Power BI are as follow:

a. Drill-Down

- Lets we explore data within a single visual by moving through hierarchical levels.
- Example: In a bar chart of sales, we can drill down from Year → Quarter → Month → Day.
- It keeps us inside the same visual, simply changing the level of detail shown.

b. Drill-Through

- Allows us to navigate from one report page to another, passing context along.
- Example: Clicking on a product in a summary chart can take us to a detailed report page showing region-wise sales for that product.
- It's about moving between pages, not just levels within one visual.

Real-World Example

Imagine a retail company analyzing sales:

- a. Drill-Down: A manager looks at total sales for 2025. By drilling down, they can see quarterly sales, then monthly sales, and finally daily sales — all within the same chart.
- b. Drill-Through: The manager clicks on "Product A" in the summary dashboard. Power BI takes them to a separate page dedicated to "Product A," showing detailed breakdowns like region-wise performance, customer segments, and profitability.

13. What is the “keep all filters” option in drill through? How does it affect user interaction.

ANS- Keep All Filters ensures that the context (filters, slicers, selections) from the original visual or page is preserved when navigating to the detailed Drill-Through page.

It Affects User Interaction in the following ways:

a. With Keep All Filters ON

- All filters applied on the source page (e.g., Region = North, Year = 2023) will also be applied to the Drill-Through page.
- This makes the detailed page consistent with the user's current view, avoiding confusion.
- Example: If we drill through on "Product A" while filtering for India, 2023, the detailed page will show only Product A's sales in India, 2023.

b. With Keep All Filters OFF

- Only the specific Drill-Through field (e.g., Product A) is passed to the target page.
- Other filters (like Region or Year) are ignored, so the detailed page may show broader data.
- Example: Drilling through on "Product A" would show all regions and years, not just the filtered ones.